Convex optimization for machine learning – TD1

Olivier Fercoq
Bruno Costacèque

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1 Exercises

Exercise 1. (Banach-Picard's fixed point theorem)

Let E be a complete metric space and $f : E \to E$ a **contracting** mapping, that is:

$$\exists \rho \in [0, 1[,\ \forall x, y \in E,\ \mathrm{d}\big(f(x), f(y)\big) \leq \rho \mathrm{d}(x, y).$$

Prove that f admits **one unique** fixed point[1] x^*, and that the sequence defined by:

$$\begin{cases} x_0 \in E \\ x_{n+1} = f(x_n),\ \forall n \in \mathbb{N}. \end{cases}$$

converges to x^* with rate of convergence ρ^n, *i.e.*:

$$\forall n \in \mathbb{N},\ \mathrm{d}(x_n, x^*) \leq \rho^n \mathrm{d}(x_0, x^*).$$

Exercise 2. (Gradient calculus)

1. Calculate the gradient of the following functions. A, M and Q are fixed matrices, and $\mathbf{b}$ is a fixed vector. More precisely, M belongs to $\mathcal{M}_{m,n}(\mathbb{R})$ and Q to $\mathcal{M}_{p,n}(\mathbb{R})$.

   (a)
   $$\begin{array}{rccc} f_1 & : & \mathbb{R}^n & \to & \mathbb{R} \\ & & \mathbf{x} & \mapsto & \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2 \end{array}$$

   (b)
   $$\begin{array}{rccc} f_2 & : & \mathbb{R}^n & \to & \mathbb{R} \\ & & \mathbf{x} & \mapsto & \sum_{i=1}^n \log(1 + \exp(x_i)) \end{array}$$

   (c)
   $$\begin{array}{rccc} f_3 & : & \mathcal{M}_{m,p}(\mathbb{R}) & \to & \mathbb{R} \\ & & P & \mapsto & \frac{1}{2}\|M - PQ\|_F^2 \end{array}$$

2. Let g_1, g_2 and g_3 be functions such that $g_1 : \mathbb{R}^{n_1} \to \mathbb{R}^{n_2}$, $g_2 : \mathbb{R}^{n_2} \to \mathbb{R}^{n_3}$ and $g_3 : \mathbb{R}^{n_3} \to \mathbb{R}$. Let:

   $$f_4 = g_3 \circ g_2 \circ g_1.$$

   Compute the gradient of f_4 using the Jacobian matrices of g_i for $i \in [\![1, 3]\!]$.

[1] A point such that $f(x^*) = x^*$.

3. Suppose that computing one element of the Jacobian matrices costs C_J and that multiplying two numbers costs C_M. How much does it cost to compute $\nabla f_4(x)$?

Exercise 3. Let f be a $\mathcal{C}^2$ function from $\mathbb{R}^d$ to $\mathbb{R}$. Assume there exist $\mu, L \in \mathbb{R}_+$ such that[2]

$$\mu I_d \preceq \nabla^2 f(\mathbf{x}) \preceq L I_d.$$

1. Let $M \in \mathcal{S}(\mathbb{R}^d)$ be a real symmetric matrix satisfying the previous assumption and note $\|\cdot\|_2$ its operator matrix with respect to the Euclidean norm on $\mathbb{R}^d$.

   (a) Let λ_1 be the greatest eigenvalue of M. Show that

   $$\|M\|_2 = |\lambda_1|.$$

   (b) Prove that:

   $$\mu \leq \|M\|_2 \leq L.$$

2. Show that the fixed point operator $F : \mathbf{x} \mapsto \mathbf{x} - \gamma \nabla f(\mathbf{x})$ is contractant for every γ in $]0, 2/L[$.

3. Show that the gradient method converges linearly[3] to some x^*.

4. How many iterations are necessary to ensure that $\|x_k - x^*\| \leq \varepsilon$?

[2] Where $\preceq$ denotes the partial order known as *Loewner order*: if M, N are positive semi-definite matrices, $M \preceq N$ if $N - M$ is positive semi-definite.

[3] That is:

$$\exists \rho \in]0,1[, \ \lim_{n \to \infty} \big(\mathrm{d}(x_{n+1}, x^*)\big)^{\frac{1}{n}} \leq \rho.$$

2 Solutions

Exercise 1 • Assume first we have proved that the sequence $(x_n)_n$ converges to some limit x^*. Then by continuity of f (which is ρ-Lipschitz by assumption), we have:

$$f(x^*) = f\big(\lim_{n\to\infty} x_n\big) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = x^*,$$

and so x^* would be a fixed point. It is unique since if there were another one, say y^*, we would have:

$$\mathrm{d}(x^*, y^*) = \mathrm{d}\big(f(x^*), f(y^*)\big) \leq \rho \mathrm{d}(x^*, y^*) < \mathrm{d}(x^*, y^*),$$

a contradiction.

- Now let us prove that $(x_n)_n$ indeed converges. So far we have no real clue about the value of its limit. But we know E is a complete metric space, and so Cauchy sequences always converge in E to some limit[4]. So we have to prove that (x_n) is a Cauchy sequence. Let $n, m \in \mathbb{N}$. We have:

$$\mathrm{d}(x_{n+m}, x_n) = \mathrm{d}\big(f^{\circ n}(x_m), f^{\circ n}(x_0)\big) \leq \rho^n \mathrm{d}(x_m, x_0).$$

Letting n go to infinity, we get that $(x_n)_n$ is indeed a Cauchy sequence in the complete space E. Using that x^* is a fixed point of f, the same reasoning yields:

$$\forall n \in \mathbb{N},\ \mathrm{d}(x^*, x_n) \leq \rho^n \mathrm{d}(x^*, x_0).$$

Remarque 1. ► *The proof of this theorem is very simple in comparison to its usefulness. For example it is the main argument behind the proof of Cauchy-Lipschitz, another powerful theorem which ensures the existence and the uniqueness of a solution for a very large class of ODEs*[5]*.*

► ***All** the hypotheses of the statement of this theorem are important. For example, if ρ is not strictly lesser than 1, troubles may appear. In particular, it is not enough to ask that*

$$\forall x, y \in E,\ \mathrm{d}\big(f(x), f(y)\big) < \mathrm{d}(x, y),$$

unless E is compact (and metric, so complete). It is a good exercise to try and prove this result.

Exercise 2 1. (a) The function f_1 can be decomposed as $f_1 = N \circ \varphi$, where $N(\mathbf{x}) = \|\mathbf{x}\|_2^2/2$ et $\varphi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$. The latter is a linear map, so its Jacobian matrix is A since:

$$\varphi(\mathbf{x} + \mathbf{h}) = A(\mathbf{x} + \mathbf{h}) = \varphi(\mathbf{x}) + A\mathbf{h}.$$

As for the Euclidean norm function, its differential can be computed directly too:

$$N(\mathbf{x} + \mathbf{h}) = N(\mathbf{x}) + \langle \mathbf{x}, \mathbf{h} \rangle + N(\mathbf{h}),$$

and so $\mathrm{d}N(\mathbf{x}).\mathbf{h} = \langle \mathbf{x}, \mathbf{h} \rangle$, thus its gradient is just $2\mathbf{x}$. Thus the chain rule gives:

$$\mathrm{d}f_1(\mathbf{x}).\mathbf{h} = \mathrm{d}N(A\mathbf{x} + \mathbf{b}).(A\mathbf{h}) = \langle A\mathbf{x} + \mathbf{b}, A\mathbf{h} \rangle = \big\langle A^\top (A\mathbf{x} + \mathbf{b}), \mathbf{h} \big\rangle,$$

which implies that:

$$\nabla f_1(\mathbf{x}) = A^\top (A\mathbf{x} + \mathbf{b}).$$

[4] The original purpose of Cauchy sequences and complete spaces is precisely to show that a given sequence converges, even without knowing a candidate for its limit! It is extremely useful to prove very general and powerful existence theorems, as you will soon see.

[5] Unfortunately this kind of result does not exist for PDEs, which is a huge source of troubles...

(b) The easiest way to explicit this gradient is to compute the partial derivatives of f_2. We find:

$$\forall i \in \mathbb{R}^n, \forall \mathbf{x} \in \mathbb{R}^n,\ \partial_i f_2(\mathbf{x}) = \frac{e^{x_i}}{1+e^{x^i}} = \frac{1}{1+e^{-x_i}} =: \sigma(x_i),$$

so the i-th partial derivative of f_2 is the **logistic function**, which is often used as an activation function for neural networks, especially in classification. Thus the gradient is:

$$\nabla f_2(\mathbf{x}) = \begin{pmatrix} \sigma(x_1) \\ \vdots \\ \sigma(x_n) \end{pmatrix}.$$

(c) Formally speaking, this computation looks exactly the same as in the first question, but this time, the argument P is on the left of the parameter matrix Q (while the argument $\mathbf{x}$ was on the right of the parameter matrix A). This time, we will develop the squared norm of $M-(P+H)Q$ divided by two to find directly the differential of f_3 at P taken in $H \in \mathcal{M}_{m,p}(\mathbb{R})$:

$$f_3(P+H) = \frac{1}{2}\|M-PQ\|_F^2 - \langle M-PQ, HQ\rangle_F + \frac{1}{2}\|HQ\|_F^2,$$

and so

$$\mathrm{d}f_3(P).H = -\langle M-PQ, HQ\rangle_F = -\mathrm{Tr}\big((M-PQ)Q^\top H^\top\big) = \big\langle (M-PQ)Q^\top, H\big\rangle_F.$$

Therefore, we get the following gradient (don't forget the minus sign!):

$$\nabla f_3(\mathbf{x}) = -(M-PQ)Q^\top.$$

Remarque 2. *When dealing with this kind of tedious (but important) computation, it is essential to do as many sanity-checks as possible. Here, you could verify that your result is well-defined by looking at the dimensions of the matrices involved: $M-PQ$ belongs to $\mathcal{M}_{m,n}(\mathbb{R})$ and $Q^\top$ belongs to $\mathcal{M}_{n,p}(\mathbb{R})$, so our gradient lives in $\mathcal{M}_{m,p}(\mathbb{R})$, which is consistent with the definition of f_3. Plus its expression bears a lot of similarities with the gradient of f_1!*

2. Applying the chain rule two times, first to $g_3 \circ g_2$ and g_1, and then to g_3 and g_2, we find:

$$\begin{aligned}
\forall \mathbf{x}, \mathbf{h} \in \mathbb{R}^{n_1},\ \mathrm{d}f_4(\mathbf{x}) &= \mathrm{d}(g_3 \circ g_2)(g_1(\mathbf{x})).\big(\mathrm{d}g_1(\mathbf{x}).\mathbf{h}\big) \\
&= \mathrm{d}g_3\big(g_2(g_1(\mathbf{x}))\big).\big(\mathrm{d}g_2(g_1(\mathbf{x})).\mathrm{d}g_1(\mathbf{x}).\mathbf{h}\big) \\
&= \mathrm{d}g_3\big((g_2 \circ g_1)(\mathbf{x})\big).\big(\mathrm{d}g_2(g_1(\mathbf{x})).\mathrm{d}g_1(\mathbf{x}).\mathbf{h}\big) \\
&= \mathrm{d}g_3\big((g_2 \circ g_1)(\mathbf{x})\big).\big(\mathrm{d}g_2(g_1(\mathbf{x})) \circ \mathrm{d}g_1(\mathbf{x})\big).\mathbf{h} \\
&= \big\langle Jg_1(\mathbf{x})^\top Jg_2(g_1(\mathbf{x}))^\top \nabla g_3\big((g_2 \circ g_1)(\mathbf{x})\big), \mathbf{h}\big\rangle,
\end{aligned}$$

and this horrible expressions reads as 'the differential of g_3 computed at point $(g_2 \circ g_1)(\mathbf{x})$ taken[6] in the differential of g_2 at $g_1(\mathbf{x})$ composed with the differential of g_1 at $\mathbf{x}$ in $\mathbf{h}$'. Notice the different organization of the parentheses between the third and the fourth lines! Finally we find:

$$\nabla f_4(\mathbf{x}) = Jg_1(\mathbf{x})^\top Jg_2(g_1(\mathbf{x}))^\top \nabla g_3\big((g_2 \circ g_1)(\mathbf{x})\big).$$

[6] As a linear mapping.

3. This question looks simple but there is a huge subtlety here. First, let's compute the cost $C_{m,n,p}$ of multiplying two matrices $A \in \mathcal{M}_{m,n}(\mathbb{R})$ and $B \in \mathcal{M}_{n,p}(\mathbb{R})$. The (i,j)-th coefficient of AB is the scalar product of the i-th line of A and the j-th column of B, so it requires mnp operations to compute all the coefficients AB and:

$$C_{m,n,p} = mnpC_M.$$

Now, if we compute the gradient of f_4 by multiplying matrices from left to right in the formula of the previous question, we must compute the product of two matrices, then multiply the result with a vector, so the multiplication cost is $C_{n_1,n_2,n_3} + C_{n_1,n_3,1} = (n_1n_2n_3 + n_1n_3)C_M$. But if we multiply from right to left, we **always** manipulate vectors and so we find instead $C_{n_2,n_3,1} + C_{n_1,n_2,1} = (n_2n_3 + n_1n_2)C_M$. This approach is much more efficient than the 'natural' way of multiplying matrices! It accounts a lot for the success of modern deep learning algorithms, even though it looks terribly simple!

Finally, using the backpropagation approach rather than the naive one, we find a total cost equal to $(n_1n_2 + n_2n_3 + n_3)C_J + (n_1n_2 + n_2n_3)C_M$.

Remarque 3. *f_1 is useful for least squares and regression problems while f_2 is useful for logistic regression and binary classification, and f_3 is useful for nonnegative matrix factorization.*

Exercise 3 1. (a) Since M is a real symmetric matrix, it is diagonalizable in an orthonormal basis:

$$\exists P \in \mathbf{O}(d),\ M = P\mathrm{diag}(\lambda_1, \dots, \lambda_d)P^\top =: PDP^\top.$$

Let $\mathbf{x}$ a unit norm vector. We have:

$$\|M\mathbf{x}\|^2 = \|PDP^\top\mathbf{x}\|^2 = \|DP^\top\mathbf{x}\|^2 = \sum_{i=1} \lambda_i^2 x_i \leq \lambda_1^2\|\mathbf{x}\|^2 = \lambda_1^2.$$

Thus $\|M\|_2 \leq |\lambda_1\|$. The equality case is attained by taking $\mathbf{x}$ the unit norm eigenvector associated with λ_1.

(b) Since the eigenvalues of $M - \mu I_d$ are the eigenvalues of M minus μ, and since the eigenvalues of a semi-definite positive matrix are positive, we have that:

$$\mathrm{Sp}(M - \mu I_d) \subseteq \mathbb{R}_+.$$

In particular, all the eigenvalues of M are greater[7] than $\mu \geq 0$, and thus:

$$\mu \leq \lambda_1 = |\lambda_1| = \|M\|_2.$$

The same line of reasoning yields $\|M\|_2 \leq L$, thus proving the desired result.

2. The gradient of $\mathbf{x} \mapsto \mathbf{x} - \gamma\nabla f(\mathbf{x})$ is $F : \mathbf{x} \mapsto I_d - \gamma\nabla^2 f(\mathbf{x})$. We must prove that its operator norm is strictly less than 1 for all $\mathbf{x}$. Since:

$$\forall \mathbf{x} \in \mathbb{R}^d,\ \mathrm{Sp}\big(F(\mathbf{x})\big) = 1 - \gamma\mathrm{Sp}\big(F(\mathbf{x})\big),$$

we know that the greatest eigenvalue λ_1 of $F(\mathbf{x})$ belongs to $[1 - \gamma L, 1 - \gamma\mu]$ by assumption. As a result:

$$\forall \mathbf{x} \in \mathbb{R}^d, -1 < 1 - \gamma L \leq \|F(\mathbf{x})\|_2 \leq 1 - \gamma\mu < 1.$$

[7] And so, the hypothesis on f implies that $\nabla^2 f(\mathbf{x})$ is semi-definite positive for all $\mathbf{x}$, *i.e.* f is convex on $\mathbb{R}^d$.

In particular[8]

$$\rho := \sup_{\mathbf{x}\in\mathbb{R}^d} \|F(\mathbf{x})\|_2 < 1,$$

and so $\mathbf{x} \mapsto \mathbf{x} - \gamma\nabla f(\mathbf{x})$ is contractant.

3. It suffices to apply the Banach-Picard's fixed point theorem to the contracting map $\mathbf{x} \mapsto \mathbf{x} - \gamma\nabla f(\mathbf{x})$.

4. Thanks to the the first exercise, we have the estimate:

$$\forall n \in \mathbb{N},\ \|\mathbf{x}_n - \mathbf{x}^*\| \leq \rho^n \|\mathbf{x}_0 - \mathbf{x}^*\|,$$

where $(x_n)_n$ is defined by the gradient method with starting point $\mathbf{x}_0$. Since:

$$\begin{aligned}\rho^n\|\mathbf{x}_0 - \mathbf{x}^*\| \leq \varepsilon &\iff n\log(\rho) + \log\left(|\mathbf{x}_0 - \mathbf{x}^*\|\right) \leq \log(\varepsilon)\\ &\iff n \geq \frac{1}{\log(\rho)}\left[\log(\varepsilon) - \log\left(\|\mathbf{x}_0 - \mathbf{x}^*\|\right)\right] =: \eta\end{aligned}$$

we need at least $\lceil\eta\rceil = \lfloor\eta\rfloor + 1$ iterations before we get an estimate of $\mathbf{x}^*$ with an error less than ε. Of course this result is not very useful in practice since we do not know $\mathbf{x}^*$. But it underlies the importance of choosing our initial guess $\mathbf{x}_0$ as closely as possible from $\mathbf{x}^*$.

[8] The terms $1-\gamma L$ and $1-\gamma\mu$ are essential, as strict inequalities become large inequalities when taking the supremum, and so we would have lost the very contraction property we have been trying to prove! For example, $1 - n^{-1} < 1$, but $\sup_{n\geq 0}(1 - n^{-1}) = 1$...